

Magnetic Field Scanner - System Characterization

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Section 0: Methodology

- **Environment:** Benchmarks captured across multiple Cycle Counts (CC) to characterize the full hardware envelope.
- **Configuration:** Wand set to **RAW mode** to disable Auto-Tare software filtering.
- **Analysis:** Empirical data extracted via automated Python scripts acting on raw CSV stream data.

Section 1: Operating Frequency & Latency

Cycle Count (CC)	Update Rate (Hz)	Latency (ms)
200	33.8	29.6
400	43.5	23.0
800	27.7	36.2
1600	16.3	61.5
3200	9.2	108.2

Section 2: Magnetic Noise Floor & Sensitivity

Cycle Count (CC)	RMS Noise Floor (μ T)	Peak-to-Peak Jitter (μ T)
200	$\pm$ 0.1943	1.1
400	$\pm$ 0.1875	0.8744

Cycle Count (CC)	RMS Noise Floor (μT)	Peak-to-Peak Jitter (μT)
800	± 0.1303	0.7005
1600	± 0.0694	0.3984
3200	± 0.0302	0.1831

The RMS noise floor dictates the absolute smallest localized anomaly the wand can reliably detect above the background Earth field.

Section 3: Gradiometer Isolation & Measurement Precision

Cycle Count (CC)	Max Target Signal (μT)	Precision Peak Variance (μT)	Earth Baseline Drift (μT)
200	86.1	± 8.7962	± 0.1234
400	71.38	± 1.3955	± 0.113
800	84.17	± 4.2943	± 0.0884
1600	85.73	± 4.9944	± 0.2221
3200	85.77	± 0.4399	± 0.0733

Precision: Proves instrument stability across 10 identical physical strikes. **Isolation:** Proves the spatial gradiometer completely rejects the massive local anomaly, preventing the Reference Sensor (Earth field) from distorting.

Section 4: Dynamic Range & Saturation

Cycle Count (CC)	Digital Clipping Limit (μT)	Physical Core Blind State (μT)
200	533.3	60.0
400	532.03	60.0
800	526.23	61.0
1600	457.23	62.0
3200	88.73	60.0

Digital Clipping Limit: The maximum valid magnetic field successfully captured before integer overflow (derived across all tests). **Physical Core Blind State:** The steady-state math output (~Earth's background) when a massive external field physically collapses the inductor core, blinding the sensor.

Section 5: Open Air Rebar (Dipole Physics)

Cycle Count (CC)	Max Dipole Spike (μT)	Destructive Null Dip (μT)
200	253.4	0.61
400	296.89	0.74
800	526.23	0.35
1600	457.23	0.83
3200	20.09	1.35

Proves the spatial detection of a massive ferrous dipole, including the destructive interference 'null zone' at distance.

Section 6: Attitude Tracking & AHRS Stability

Cycle Count (CC)	Max Pitch Tumble (deg)	Compass Azimuth Drift (deg)
200	126.8	± 16.68
400	129.1	± 21.98

Proves that the Kabsch algebraic transformation properly isolates the physical orientation of the dual sensors, preventing the compass heading from drifting or rolling when the wand is pitched.

Section 7: Calibration Consistency

(Analyzed 6 discrete calibration logs)

Proves the mathematical repeatability of the figure-8 calibration routine by ensuring the Kabsch algorithm converges on statistically identical hard/soft iron matrices across multiple runs.

Section 8: In-Wand Math Processing Power

- **Algorithm:** 9-parameter Least-Squares Ellipsoid Fit + Kabsch Rotational Alignment
- **Calibration Matrix Generation Time:** ~10 - 12 seconds

Real-time Vector Correction Time: < 1 ms

The ESP32-S3 successfully computes the massive matrix inversion and eigen-decomposition on 1,200 floating-point 3D vectors in ~12 seconds after tumbling, and then applies that 9-parameter matrix to real-time streams at over 1,000 Hz.